What Problems Did Bitcoin Solve?

Translate

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What makes Bitcoin so revolutionary?

Why is it often hailed as a game-changer in the world of money?

As mentioned in an earlier lesson, a mysterious figure who called himself, Satoshi Nakamoto authored a white paper titled, Bitcoin: A Peer-to-Peer

The paper revealed details of creating “electronic” cash” (digital currency) free of control from any organization or government.

During Satoshi Nakamoto’s research though, he discovered there have been **multiple attempts** in the past to create a digital currency.

There were early pioneers such as b-money, Bit Gold, ecash, E-gold, Hashcash, Liberty Reserve, and RPOW. But they didn’t work out for two main reasons:

1. **Centralization**. These digital currencies were controlled by a central authority which introduced a single point of failure. Having centralized control also created other types of risk. An entity in control could decide to do something shady like create secret amounts of additional money for personal use. Or the system gets hacked and users’ money gets stolen, or the government forces the entity to shut down which means everybody’s money is now worthless.
2. **Double Spending**. There wasn’t a foolproof way to know whether the currency was being duplicated or double-spent (the digital version of counterfeiting money).

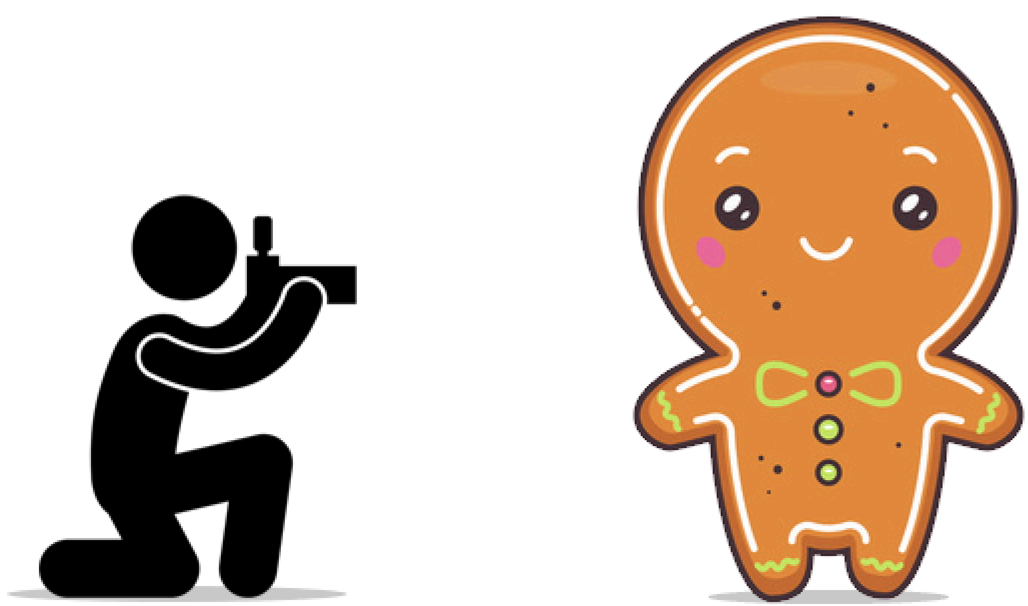
So even before Bitcoin was a twinkle in Satoshi’s eye, several attempts have been made to create *decentralized* **electronic money**, or **digital cash**, in the past, but they all failed. 😔

Unlike a photo, PDF, or other documents. you can't simply attach some money to an email and send it to someone.

Why?

Because whenever you do a transfer of VALUE between two people, you need to make sure that **a real transfer has taken place**.

For example, let's say you ran into the Gingerbread Man and were able to get a photo of it.



If I send the original digital photo to you, I can simply attach the photo to a text message and send it.

You will receive the photo.

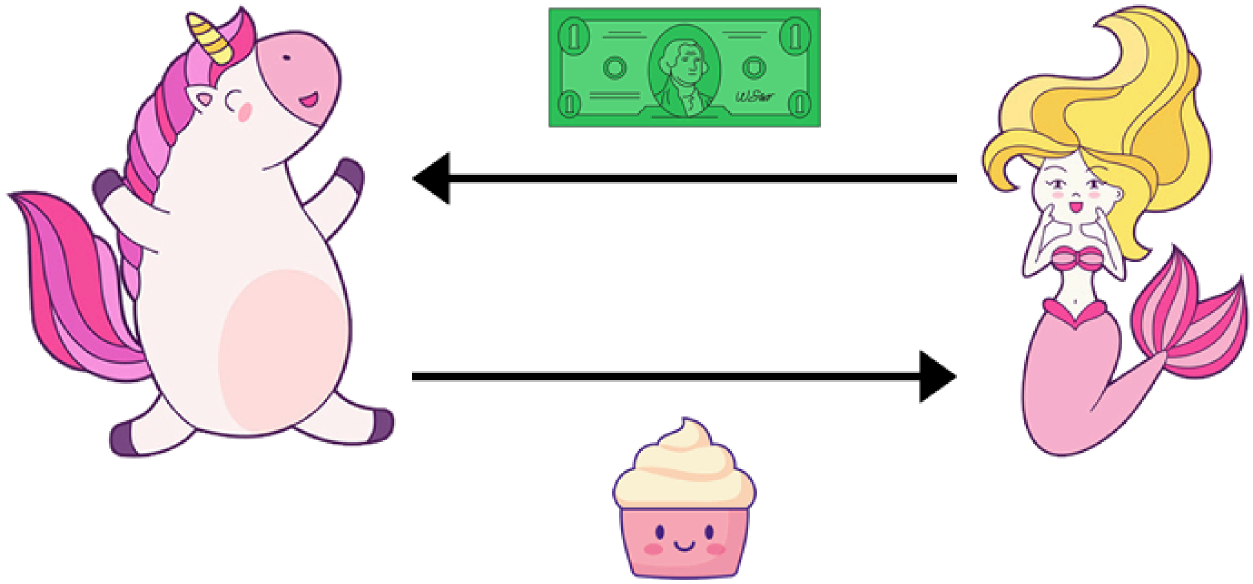
But now, there are **TWO copies of the photo**. The one attached to the text message and the original file that I have stored on my computer.

What has happened is that I've sent you a COPY of the file of the photo, not the original file.

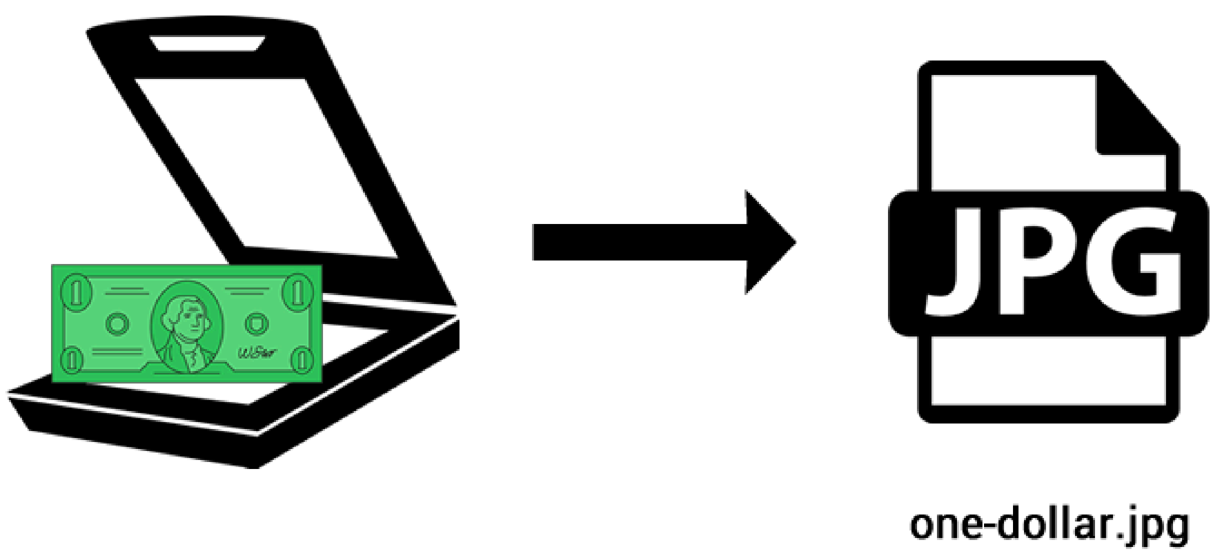
When it comes to sending digital photos, this may not be a big deal (unless you're a celebrity who loves to take nude selfies and your phone gets hacked).

But when it comes to sending digital MONEY, this is a very big deal.

Let's return to the cash transaction example involving Molly the Mermaid and Ursula the Unicorn from the previous lesson.



Imagine if Molly scanned her one-dollar bill and named the digital image, "**one-dollar.jpg**".

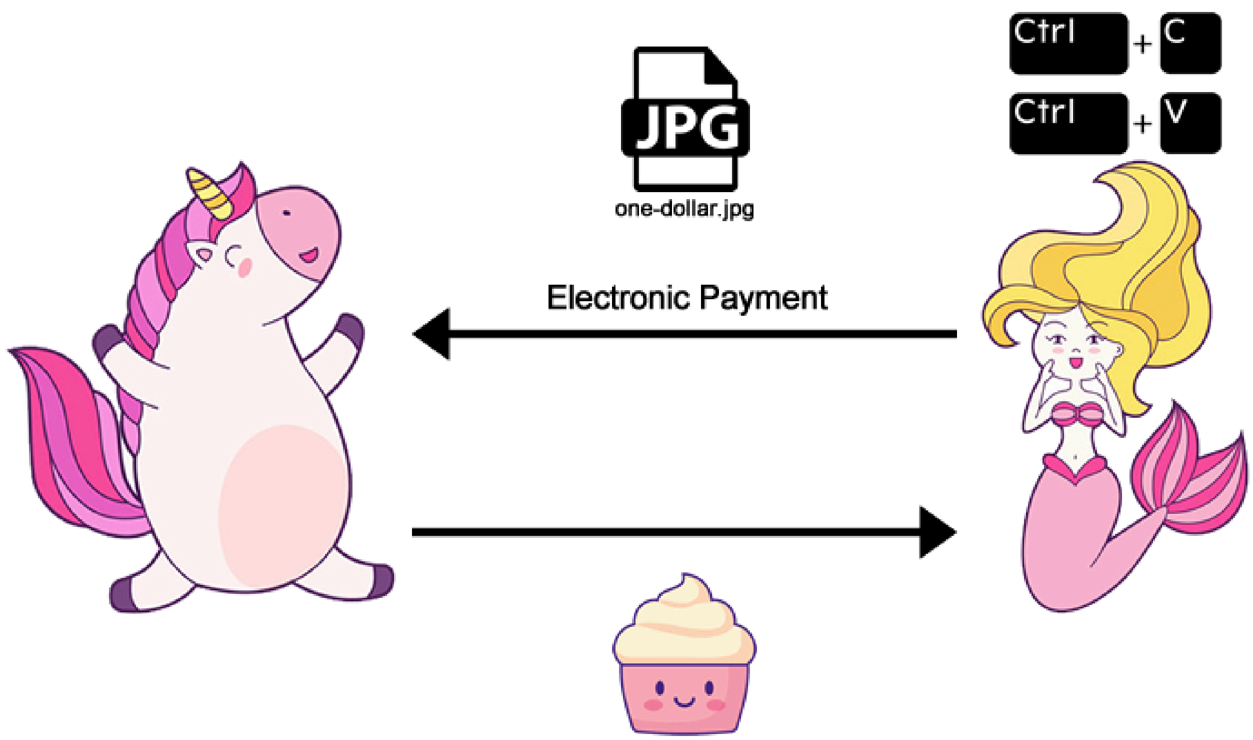


When something is in digital form, it's **easy to copy** and **duplicate** it – as

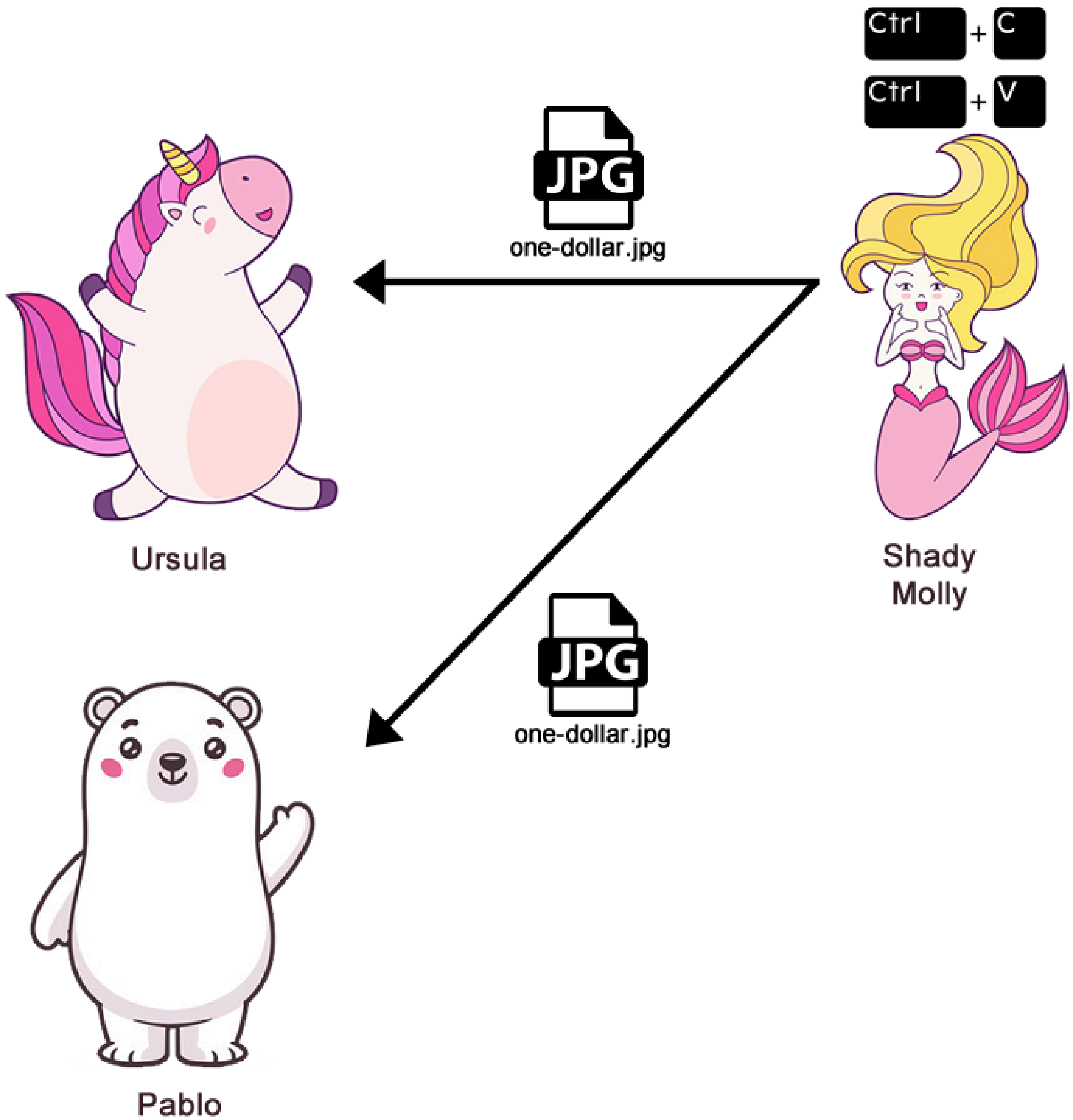
Where is the value in that?!

Molly can create an infinite number of digital copies of “one-dollar.jpg” and spend it as many times as she wants.

This issue is known as the” **double spend” problem**.



And if multiple people own the same exact “one-dollar.jpg” image file, then….w**ho is the actual owner**?



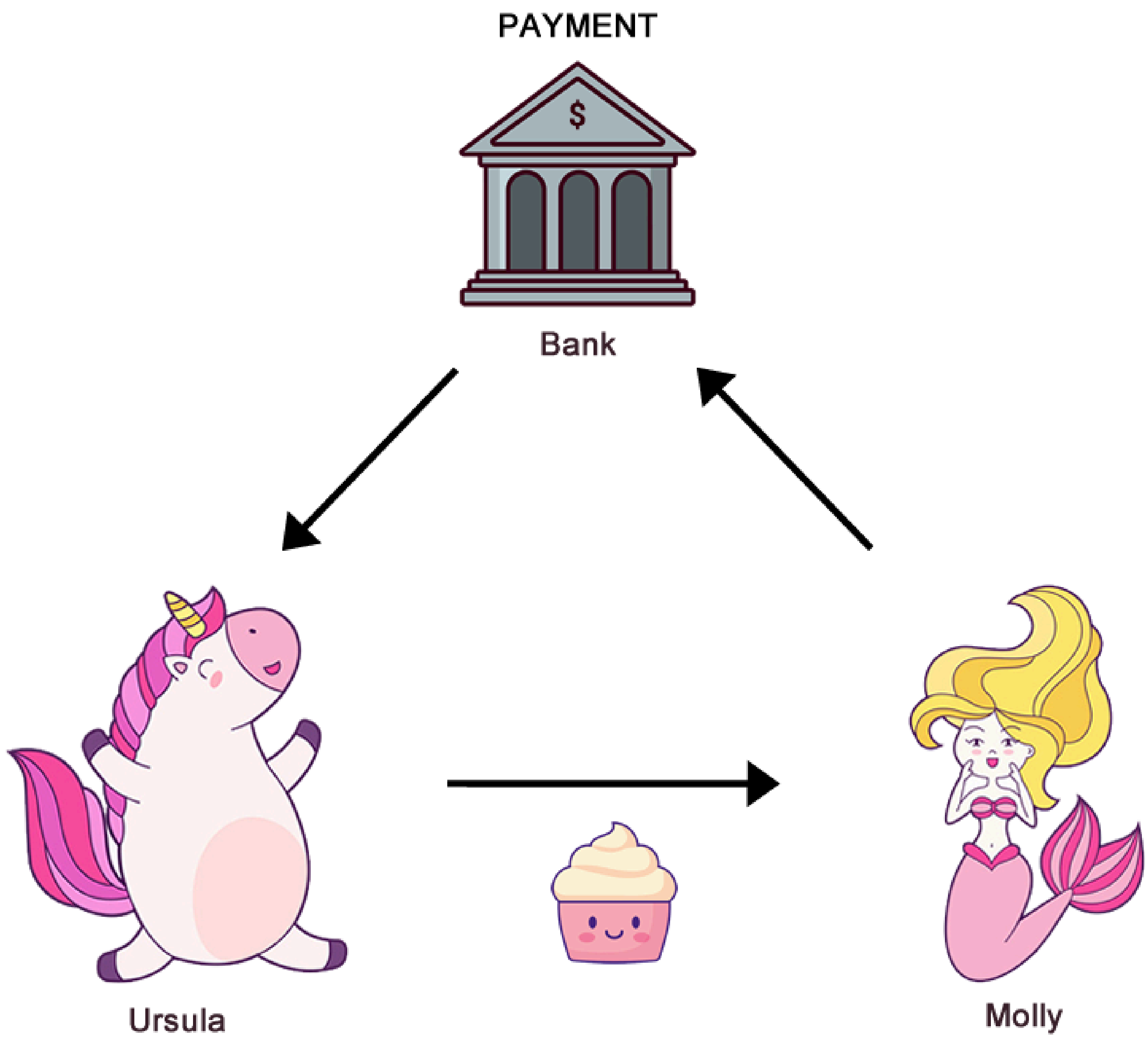
If you're trying to spend money digitally, how can you prove that the money transferred is really gone from its original place? That there was an actual **change in ownership**?

If Molly gives Ursula $1, how can Molly prove that she wasn't then going to give the **same $1** to somebody else?

When it comes to digital payments, the **net value** of all transfers needs to be equal to $0. For example, when Ursula sends $1 to Molly, Ursula should *lose* $1, and Molly should *gain* $1.

Before Bitcoin, the only way to send money electronically was through a bank or a payment company like PayPal. (And as mobile phone usage grew, fintech mobile apps like Venmo or M-Pesa.)

Basically, we had to rely on a **central authority**.



Here's what happens:

Molly starts with **$1** in her account, and Ursula has **$0**.

Molly tells the bank to transfer $1 to Ursula.

The bank adjusts Molly's and Ursula's account balances.

Molly's and Ursula's account balances exist as numbers on a computer.

Molly and Ursula trust the bank to keep their bank account balances up-to-date and accurate.

If you noticed, there is **no physical cash changing hands**.

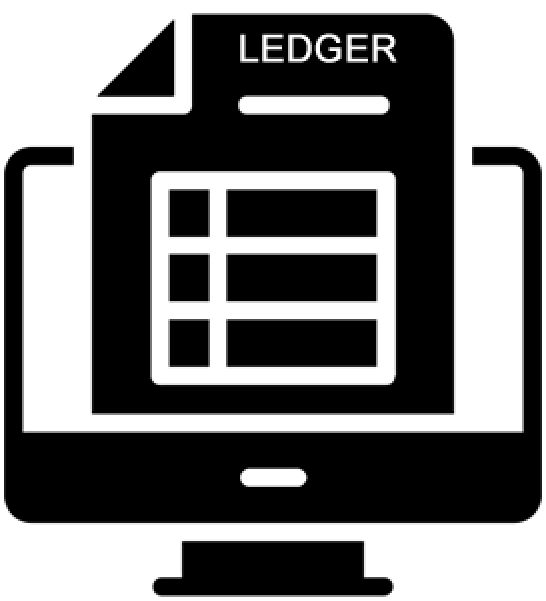
Instead of the physical $1 bill, Ursula and Molly rely on the bank to do a digital transaction **on their behalf**.

What is a ledger?

The bank keeps track of the accounts of both buyer and seller.

How does a bank keep track of its account balances?

The bank uses a **ledger**.



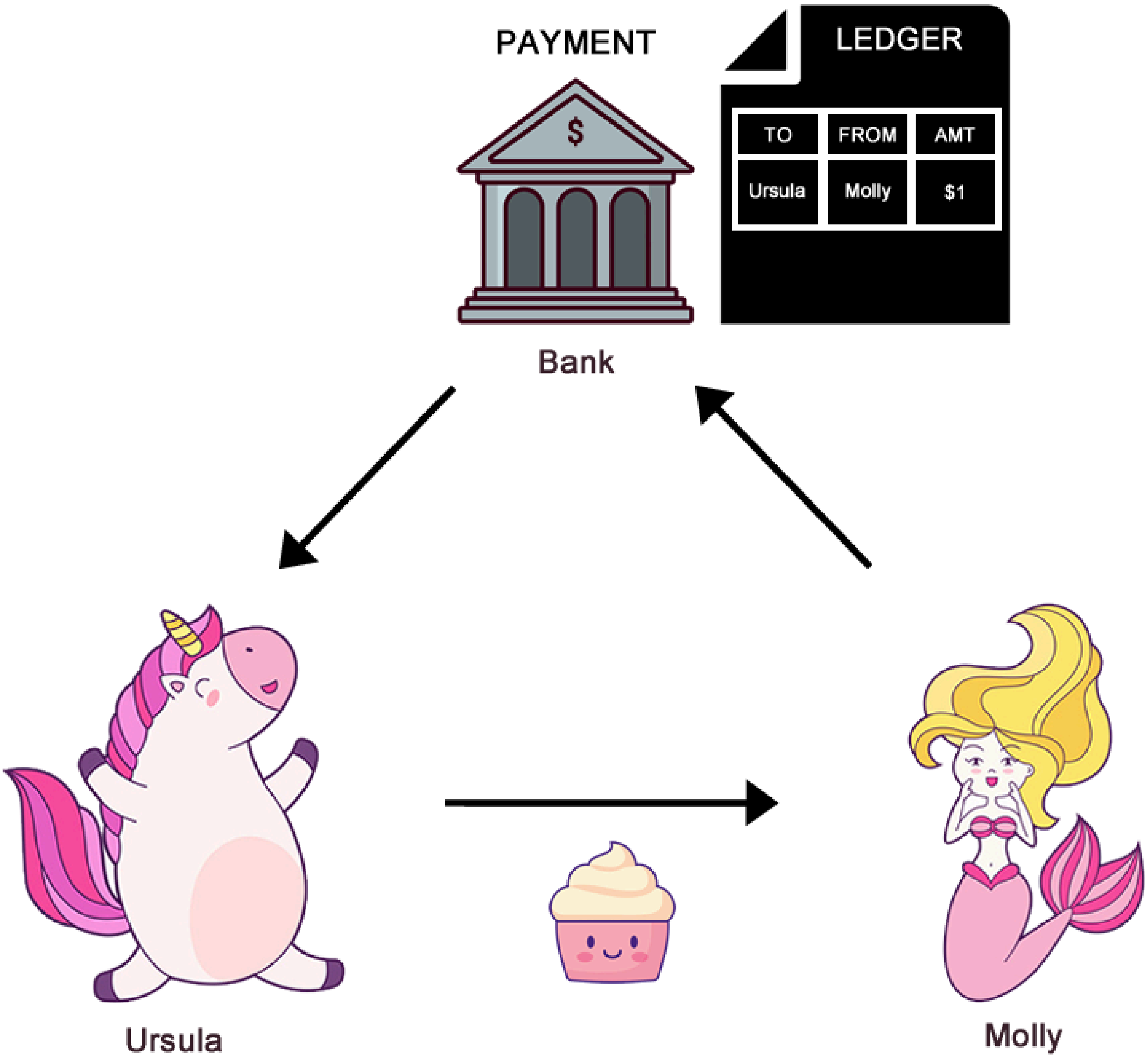
1. The ledger serves as a method for **proving ownership**, which is done by reading historical data preserved in the ledger.
2. The ledger documents any **transfer of ownership**, which means that new data is produced and written to the ledger.

In a traditional bank transaction, when Molly sends a payment to Ursula, the central authority (the bank) looks at the ledger to make sure that Molly has the funds and then decreases Molly's bank account balance and increases Ursula's bank account balance.

Basically, for electronic (non-physical) transfers of money, you need a **recordkeeping system**.

That's what the ledger does. It keeps a **record of transactions.**

In this example, the ledger recorded that Molly transferred $1 (unit of currency) to Ursula. And now Molly has $0.



This prevents the double spending problem.

For online transactions, without these intermediaries, we could theoretically just "copy and paste" money and it'd be impossible to know which transactions were legit or fraudulent.

While relying on central authorities or intermediaries solves the "double spending" problem, this requires you to **TRUST** them.

> **Double spending i**s the process of making two payments with the same funds in order to deceive the recipient of those funds. With physical currency, this isn't possible. You can't give two people the same $5 bill With online payments, you must trust a third-party to make sure funds are sent and received properly. Banks, credit card companies, and payment processors validate the transactions themselves and minimize the risk of double spending.

For example, you must trust that your bank maintains your account balances

Your transaction could also potentially be "**censored**" by the government, where the bank is pressured to block or reject transactions for political or other reasons.

Having to rely on and trust banks and other third parties are known as the **centralization problem**.

So to quickly summarize, we currently have two problems with digital money:

PROBLEM #1: Double Spending
The risk of someone being able to spend digital money twice (or more) since anything in digital format can be easily duplicated. This new amount of currency that didn't previously exist is also known as counterfeit or fraudulent money.

PROBLEM #2: Centralization
In order to solve the "double spending" problem, you have to rely on a third party to maintain and keep track of the change in ownership of funds. But this exposes you to the risk of your digital money. being stolen, confiscated, frozen, or blocked by the third party. You lose control or ownership of your money.

Satoshi Nakamoto figured out how to solve both problems!

Allowing money to move the way text messages or emails do between any two people and without any central intermediary **requires a unique solution**.

Satoshi's solution created a NEW way to use money in a digital form that is counterfeit-proof and can be sent directly from one person to another ("peer-to-peer") without having to go through a financial institution.

No more banks or other intermediaries. No more need to ask permission or get approval from them if you want to transfer money.

But how?

How the heck do you prevent double spending of digital money if you do NOT

Who then maintains the accuracy of the ledger?

Well, if you don't want to rely on traditional financial institutions, you have to start from scratch and create **a totally new SYSTEM**.

If you want to skip the bank entirely, you need a new system for tracking value and the transferring of value from one person to the next.

Not wanting to rely on banks or governments means **you can NOT be part of any existing financial system**.

Why?

Because the money used in existing financial systems is based on conventional currency, also known as "**fiat money**" such as dollars, euros, yen, pounds, and pesos. And all of these currencies are controlled by their governments which means they are all CENTRALIZED.

That's exactly what Satoshi Nakamoto wanted to avoid. He didn't want to rely on a central authority or administrator to manage the ledger.

This means that the system needed to be able to be operated by anyone, without the need to gain permission from some kind of gatekeeper.

The Bitcoin God wanted to use digital money that is DECENTRALIZED.

It would be able to operate outside of any government regulation and central banks, which means it is not under the control of any single person or organization.

It would be **global, state-free money**.

This would allow anyone to make online payments to anyone, anywhere in the world at any time.

If this money had a slogan, it would be this:

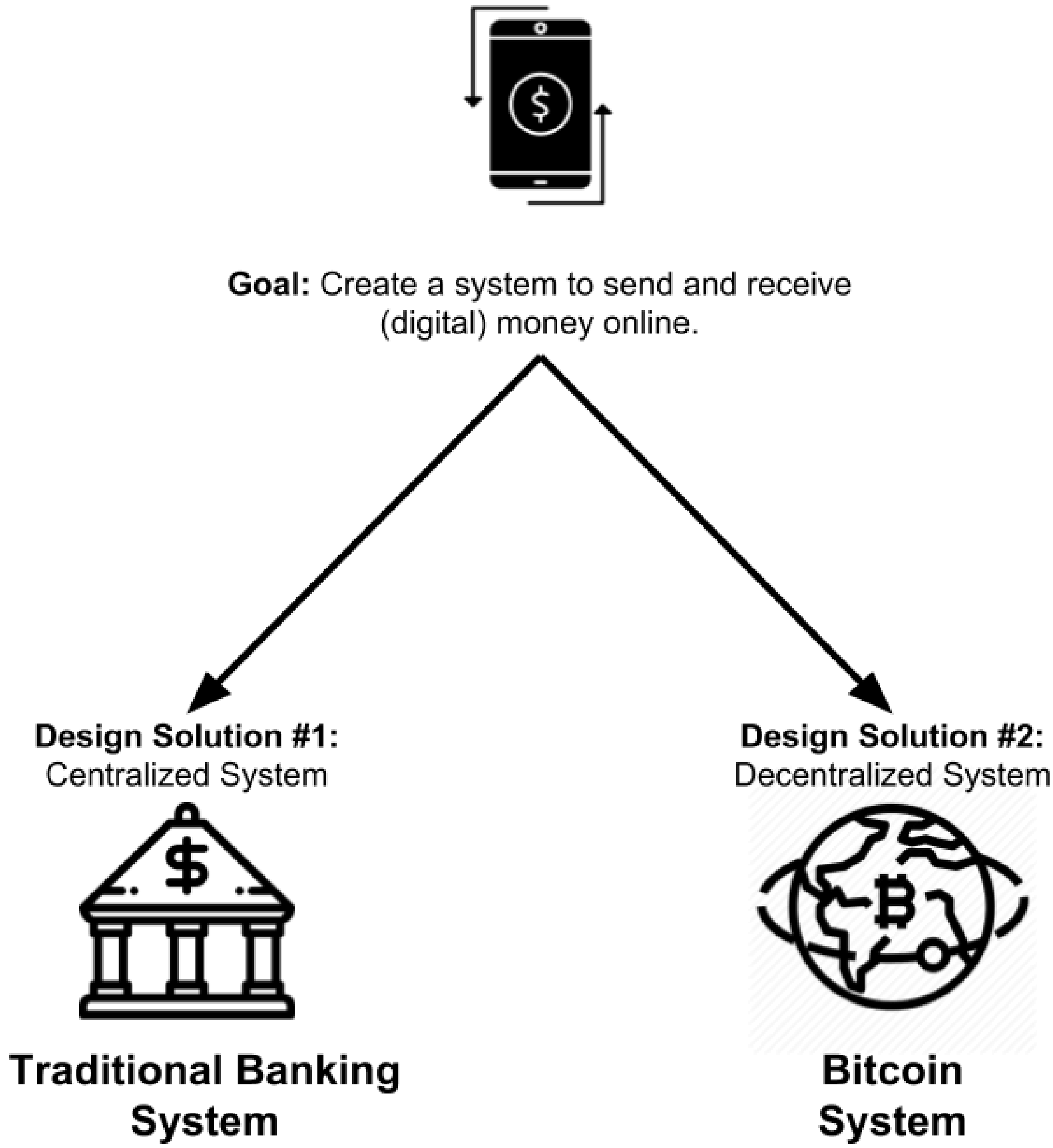
> *“Regardless of where you live in the world, you can spend your money **whenever** you want on **whatever** you want with **whomever** you want.”*

Nobody, no company, no authority, no government could stop the transaction. (This is also known as being“**censorship-resistant**”.)

Satoshi Nakamoto basically wanted an online currency for the internet that would function just like physical cash (“digital cash”), that could NOT be controlled by anyone.

So what did he do?

He went to work to create a brand new system that would do just that.

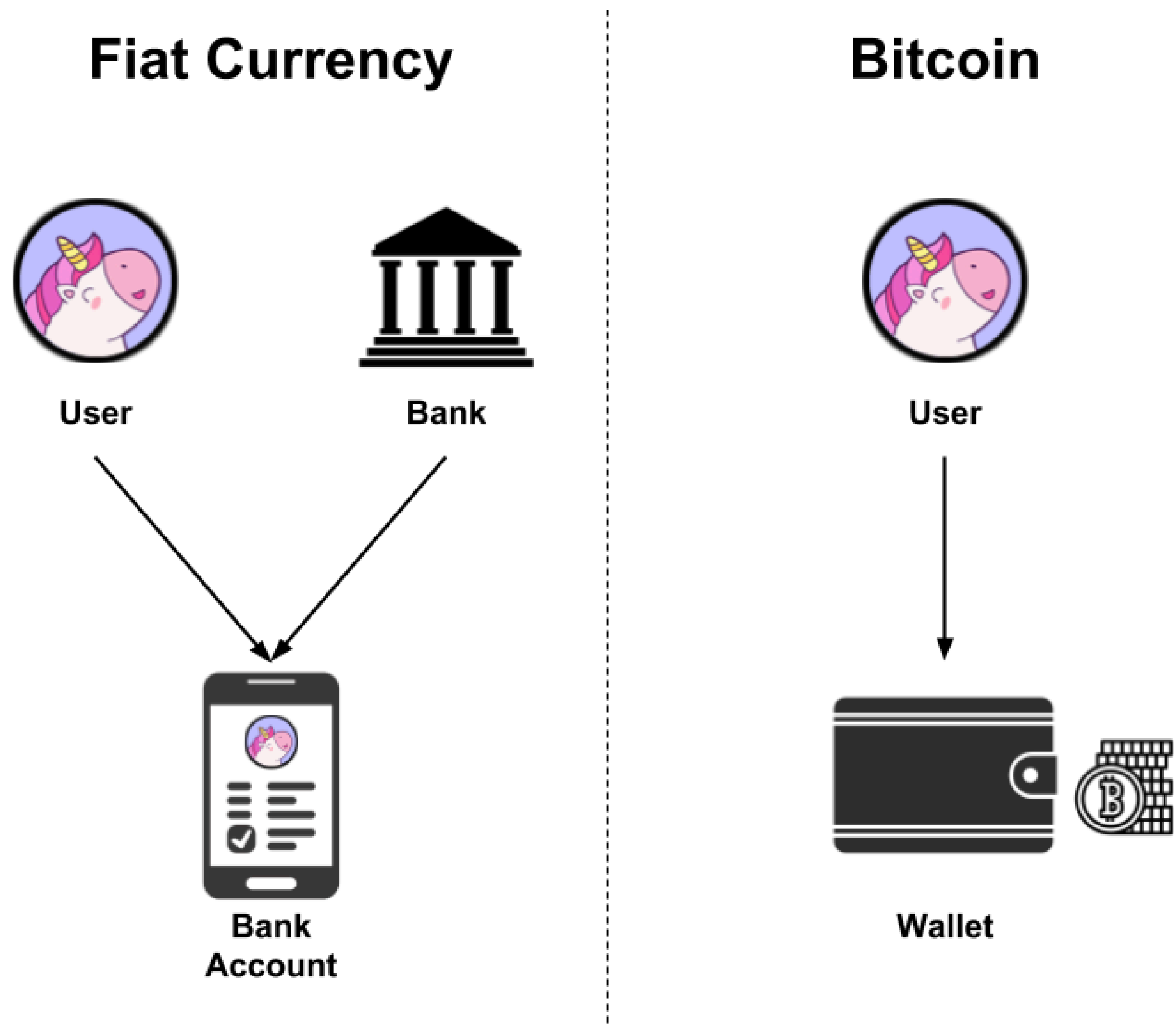


This new system would be designed to **manage the ownership and creation of its own unit of currency**.

This new SYSTEM would allow anybody with an internet connection to send, receive, and store this "digital currency."

This new system would be an **alternative** to the traditional financial system, which is built around banks.

Since there are no banks, you would not need a bank account. All you would need is a "**wallet**" which anyone can create. (I'll discuss wallets in a later lesson.)



Although many of the concepts and technologies underlying Bitcoin already existed in 2008, no one had ever **put all the pieces together**.

original way.

This new system was basically a successful Frankenstein of different technical innovations he borrowed from earlier attempts at cryptocurrencies and electronic cash in the decades before Bitcoin was launched.

Satoshi Nakamoto named this system, **Bitcoin**.

Bitcoin vs. bitcoin?

Bitcoin (capital “B”) is the system that automatically manages the ownership and creation of its own digital units of currency called **bitcoins** (lowercase “b”).

As a Bitcoin user, you’d say that you have a certain amount of “bitcoins”, similar to how you’d say you have a certain amount of British “pounds”, Nigerian “naira”, Indian “rupees”, or U.S. “dollars”.

“**BTC**” has been the generally accepted currency code for bitcoin. So **1 bitcoin = 1 BTC**.

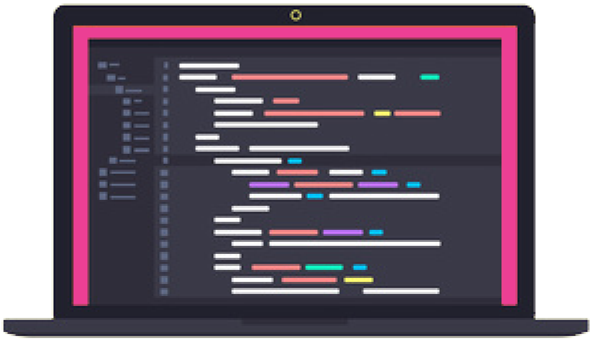
One bitcoin is divisible to eight decimal places. So it’s possible to own **0.00000001 BTC**.

Think of “Bitcoin” (capital “B”) as the **brand name** for the new system (or concept) that Satoshi Nakamoto created to move “money” around on the internet. And that “money” is denominated NOT in “dollars” or “euros” but in its own unit of account called “bitcoins”.

Why he would name the currency (lowercase “b”) as the same name as the system itself makes it obvious that he never worked in marketing. But at least now YOU know the difference!

> There are several components that make the whole Bitcoin system work, so whenever I’m talking about a specific component of the Bitcoin system, I will use “**Bitcoin**” (capital “B”). For example, “**Bitcoin wallet**”.

The entire Bitcoin system is run by **software** that Satoshi Nakamoto created.



The Bitcoin system creates bitcoins and keeps track of the change of ownership of bitcoins.

Let's imagine that when a $1 bill is freshly printed, the Federal Reserve, the U.S. central bank, starts **tracking its change of ownership.**

Whenever the $1 bill changes hands from person to person, the Federal Reserve records this on a file. The dollar bill's **entire history of ownership** is constantly tracked in sequential order from its creation.

This is basically what the Bitcoin system does. But instead of creating and tracking U.S. dollars (USD), it c**reates and tracks bitcoins (BTC)**, its own unit of account.

If you send bitcoin to someone, that transaction becomes an official entry in a file that's automatically and permanently recorded, so that bitcoin can't be spent twice.

The file stores all past transactions permanently so that there is a **complete historical trail of ownership**. This is very powerful since it proves who the current owner is without needing a third party.

And there is NO central authority, like the Federal Reserve, storing and maintaining the file.

Instead, it is stored publicly by a **network of computers** across the world. This file is replicated and stored on thousands of independent computers and is constantly updated whenever bitcoins change owners.

Every time a transaction occurs, it is **batched** together with other transactions, and every couple of minutes, the ledger is updated on every computer across the network.

This means that there are *thousands* of identical copies of this file.

Whenever bitcoins change ownership, the transaction is recorded on EVERY one of the thousands of copies of the file around the world.

The Bitcoin system is constantly comparing all copies of the file to make sure they all have matching transactions. This ensures that all copies are kept in sync.

This “file” is **Bitcoin’s ledger**.

If you think about it, **ANY system to keep track of digital money is just a RECORDKEEPING system**.

The way banks works….keeping a *central* ledger…is one type of recordkeeping system.

> When a single entity (like a bank) has control of all financial records, it is considered to be a **central ledger**.

Bitcoin is a brand new type of recordkeeping system!

Satoshi Nakamoto’s design is what allows Bitcoin to function as a recordkeeping system SEPARATE from the banks and operate totally OUTSIDE the traditional financial system.

Bitcoin does not use a central ledger. Bitcoin’s ledger is a different type of ledger known as a **distributed ledger.**

Yes No

Next Lesson
What is a Distributed Ledger?

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